E-commerce Business Performance & Customer Intelligence

A SQL Server analysis of the Olist Brazilian marketplace dataset

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96,211 Completed orders	**93,104** Customers	**BRL 13.18M** Item revenue
BRL 137.00 Average order value	**3.00%** Repeat customers	**93.21%** On-time delivery

Executive summary

This report answers 15 business questions about an online marketplace using SQL Server only: from loading and validating nine raw tables, through a clear definition of a completed sale, to sales, product, customer and delivery analysis and three evidence-based recommendations. The window analysed is January 2017 to August 2018.

- **Growth has stalled.** Revenue peaked in November 2017 (BRL 987,765) and quarterly growth fell from +77.5% to +3.8%; monthly revenue has been flat at BRL 0.84M to 0.98M since March 2018.
- **Revenue is concentrated.** SP, RJ and MG generate 63.4% of revenue and ten categories generate 62.47%. SP has the highest revenue and the lowest average order value (BRL 125.12).
- **Customers rarely return.** 97.00% bought once; 3.00% repeat. 14,053 lapsed high spenders hold 29.56% of revenue.
- **Late deliveries hurt reviews.** 93.21% of orders are on time, but late orders average 2.27 stars vs 4.29, and three months produced about 48% of all late orders.
- **Three big categories have review problems:** bed_bath_table, office_furniture and telephony hold 12.1% of revenue.

Methodology and definitions

Every question starts from the same views (v_qual_orders, v_qual_items, v_qual_order_rev), so all answers reconcile: quarterly revenue, state revenue and RFM segment revenue each sum to BRL 13,181,027.13.

Definitions used throughout

Term	Definition	Why
Completed (qualifying) sale	order_status = 'delivered', at least one item row, purchased Jan 2017 to Aug 2018	Undelivered orders can still be cancelled; delivered-only keeps revenue, reviews and delivery on one population.
Analysis window	Jan 2017 to Aug 2018	2016 has 329 orders (with a gap in November) and Sep-Oct 2018 only 20, which would distort growth rates.
Item revenue	SUM(order_items.price), freight excluded	payment_value includes instalment interest and vouchers, so it does not reconcile to item prices.
Unit	one row in order_items	The dataset has no quantity column.
Customer	customer_unique_id	customer_id changes with every order: 99,441 ids vs 96,096 real customers.
Late delivery	delivered DATE after estimated DATE	The estimate is a date, so comparing timestamps would overstate lateness.
Missing values	kept as NULL, never replaced with 0	A missing delivery date means "not delivered", not "delivered instantly".

Part A: Data quality and definitions

Q1. Overall quality of the raw dataset

Question: What is the overall quality of the raw dataset? Count rows, find missing values, check key uniqueness and duplicates, and find invalid dates, prices or inconsistent statuses.

Answer: All nine tables loaded in full. Keys are unique everywhere except reviews, there are no invalid prices, payments or review scores, and every issue found is explainable and handled without affecting the analysis.

Row counts

Table	Rows
customers	99,441
geolocation	1,000,163
orders	99,441
order_items	112,650
order_payments	103,886
order_reviews	99,224
products	32,951
sellers	3,095
product_category_name_translation	73 (after adding 2 missing translations)

Missing values in important fields

Field	NULL rows	NULL %	Treatment
order_reviews.review_comment_message	58,256	58.71	Expected; only review_score is used
orders.order_delivered_customer_date	2,965	2.98	2,957 explained by order status (kept NULL); 8 delivered anomalies excluded from delivery metrics
products.product_category_name	610	1.85	Labelled 'unknown'
orders.order_delivered_carrier_date	1,783	1.79	Not used
orders.order_approved_at	160	0.16	Not used
products.product_weight_g	2	0.01	Not used
estimated date, customer id, price, freight, payment value, review score	0	0.00	None needed

Key uniqueness and duplicates

Key checked	Total rows	Distinct keys	Duplicates
orders (order_id)	99,441	99,441	0
customers (customer_id)	99,441	99,441	0
products (product_id)	32,951	32,951	0
sellers (seller_id)	3,095	3,095	0
order_items (order_id, order_item_id)	112,650	112,650	0
order_payments (order_id, payment_sequential)	103,886	103,886	0
order_reviews (review_id), not a clean key	99,224	98,410	814
order_reviews (order_id), orders with more than one review	99,224	98,673	551
translation (category name)	73	73	0

Invalid or inconsistent values (rows affected)

Check	Rows	Treatment
carrier pickup before approval	1,359	Flagged, not deleted; field unused
orders with no items at all	775	Excluded from sales (no basket, no sale)
customer delivery before carrier pickup	23	Flagged, not deleted; field unused
payments with value 0	9	Immaterial; revenue uses item price
status delivered but no delivery date	8	Stay in sales; excluded from delivery metrics
status canceled but has delivery date	6	Excluded from sales (status is canceled)
products with weight <= 0	4	Immaterial
payments with installments < 1	2	Immaterial
orders with no payment record	1	Excluded from payment analysis
all other checks (18 in total): delivered/approved/estimate before purchase, score outside 1-5, price <= 0, freight < 0, payment < 0, in-progress with delivery date	0	None needed

Findings

- **Geolocation** has 1,000,163 rows for only 19,015 zip prefixes, and 261,831 rows (26.2%) are exact duplicates. No question needs it, because the customer state comes from `customers`.
- **customer_id vs customer_unique_id:** 99,441 vs 96,096. Every customer question uses `customer_unique_id`.
- **Reviews:** 551 orders have more than one review, and 768 orders have none. Reviews are deduplicated to the latest per order and always LEFT JOINed.
- **Payments do not reconcile to item totals** (differences in both directions, from instalment interest and vouchers), so revenue is defined as item `price`.
- Date range: 2016-09-04 to 2018-10-17. Every order has an estimated delivery date, and there are 0 orders delivered or approved before purchase.

Approach: Row counts with UNION ALL; NULL profiling with SUM(CASE WHEN ... IS NULL); key checks with COUNT(*) minus COUNT(DISTINCT); orphan-key checks with LEFT JOIN ... IS NULL (all 0); range and date-logic checks.

Q2. Share of orders by status

Question: What percentage of orders fall into each order status?

Answer: 97.02% of orders were delivered. Canceled (0.63%) and unavailable (0.61%) orders together are 1.24%, and 1.74% were still in progress when the data was captured.

Every status present

Order status	Orders	% of orders
delivered	96,478	97.02
shipped	1,107	1.11
canceled	625	0.63
unavailable	609	0.61
invoiced	314	0.32
processing	301	0.30
created	5	0.01
approved	2	0.00

Grouped

Status group	Orders	% of orders
delivered	96,478	97.02
other (shipped, invoiced, processing, created, approved)	1,729	1.74
canceled	625	0.63
unavailable	609	0.61

Findings

- Fulfilment is very reliable, so the main operational question is delivery timing (Q14), not failed orders.

Approach: COUNT(*) per status with the share calculated as COUNT(*) / SUM(COUNT(*)) OVER ().

Q3. Definition of a completed sale

Question: How should a completed sale be defined? Document which statuses are included or excluded, how cancelled or undelivered orders are treated, and why a missing value should not automatically become zero.

Answer: A completed sale is an order with status 'delivered', at least one item row, purchased between January 2017 and August 2018. This gives 96,211 qualifying orders and leaves out only 2.7% of all item revenue.

Funnel from all orders to qualifying orders

Step	Orders
1. All orders	99,441
2. status = delivered	96,478
3. ...and has at least one item	96,478
4. ...and purchased Jan 2017 to Aug 2018 (QUALIFYING)	96,211

Item revenue by status (what the definition leaves out)

Status	Orders with items	Item revenue (BRL)	% of total	Completed sale?
delivered	96,478	13,221,498.11	97.28	Yes
shipped	1,106	150,727.44	1.11	No
canceled	461	95,235.27	0.70	No
invoiced	312	61,526.37	0.45	No
processing	301	60,439.22	0.44	No
unavailable	6	2,007.69	0.01	No
approved	2	209.60	0.00	No
Total	**98,666**	**13,591,643.70**	**100.00**	

Why NULL must not become 0

Calculation	Average delivery days
NULLs ignored (correct)	12.50
NULLs wrongly set to 0	12.12

Findings

- **Included:** delivered orders only. In-progress orders (shipped, invoiced, processing, created, approved) have not completed and can still be cancelled.
- **Excluded:** canceled and unavailable orders. 603 of the 609 unavailable and 164 of the 625 canceled orders have no items at all, so they never had a basket.
- **Revenue left out by the definition:** BRL 370,145.59, which is 2.72% of the BRL 13.59M held in all statuses.
- **The 267 delivered orders outside the window** are dropped only for trend reasons (2016 and Sep-Oct 2018 are too thin).
- **Why not zero:** a missing delivery date means the event has not happened or was not recorded. Setting it to 0 says "delivered instantly": the average falls from 12.50 to 12.12 days, and the 2,965 orders (2.98%) would count as ultra-fast, on-time deliveries. A missing review comment or product weight is likewise unknown, not zero. Keeping NULLs lets aggregates ignore them and keeps denominators honest.

Approach: Views v_orders_clean, v_qual_orders, v_qual_items and v_qual_order_rev implement the definition once, so every later query starts from the same population.

Key SQL

```
-- A completed sale: delivered, has items, purchased Jan 2017 - Aug 2018
CREATE OR ALTER VIEW dbo.v_qual_orders AS
SELECT oc.*,
       CASE WHEN oc.is_delivered_valid = 1
            THEN CASE WHEN CAST(oc.order_delivered_customer_date AS DATE)
                           > CAST(oc.order_estimated_delivery_date AS DATE)
                      THEN 1 ELSE 0 END
       END AS is_late_day
FROM dbo.v_orders_clean oc
WHERE oc.order_status = 'delivered'
  AND oc.is_revenue_order = 1
  AND oc.in_window = 1;
```

Part B: Sales and revenue performance

Q4. Overall sales KPIs

Question: What are the overall sales KPIs? Calculate distinct orders, distinct customers, item revenue, average order value (AOV) and total units sold, and define each metric and its denominator.

Answer: From January 2017 to August 2018 the marketplace completed 96,211 orders for 93,104 customers, generating BRL 13.18M in item revenue at an average order value of BRL 137.00.

Sales KPIs

KPI	Value	Definition / denominator
Orders	96,211	Distinct qualifying order_id
Customers	93,104	Distinct customer_unique_id
Item revenue	BRL 13,181,027.13	SUM(price), freight excluded
Average order value	BRL 137.00	Item revenue / distinct orders
Units sold	109,880	Rows in order_items
Units per order	1.14	Units / orders
Revenue per customer	BRL 141.57	Item revenue / distinct customers
Freight collected	BRL 2,192,092.88	Reported separately from revenue

Findings

- **Single-item baskets:** 1.14 units per order means almost every order has one item.
- **Little repeat buying:** revenue per customer (141.57) is only 3.3% above AOV (137.00).
- **Freight** adds BRL 22.78 per order, 16.6% on top of item revenue.

Approach: One aggregate query on v_qual_items with COUNT(DISTINCT ...), SUM() and CAST(... AS DECIMAL).

Q5. Best and weakest month and quarter

Question: Which month and quarter generated the highest item revenue? Show revenue by month and quarter and identify the best and weakest periods.

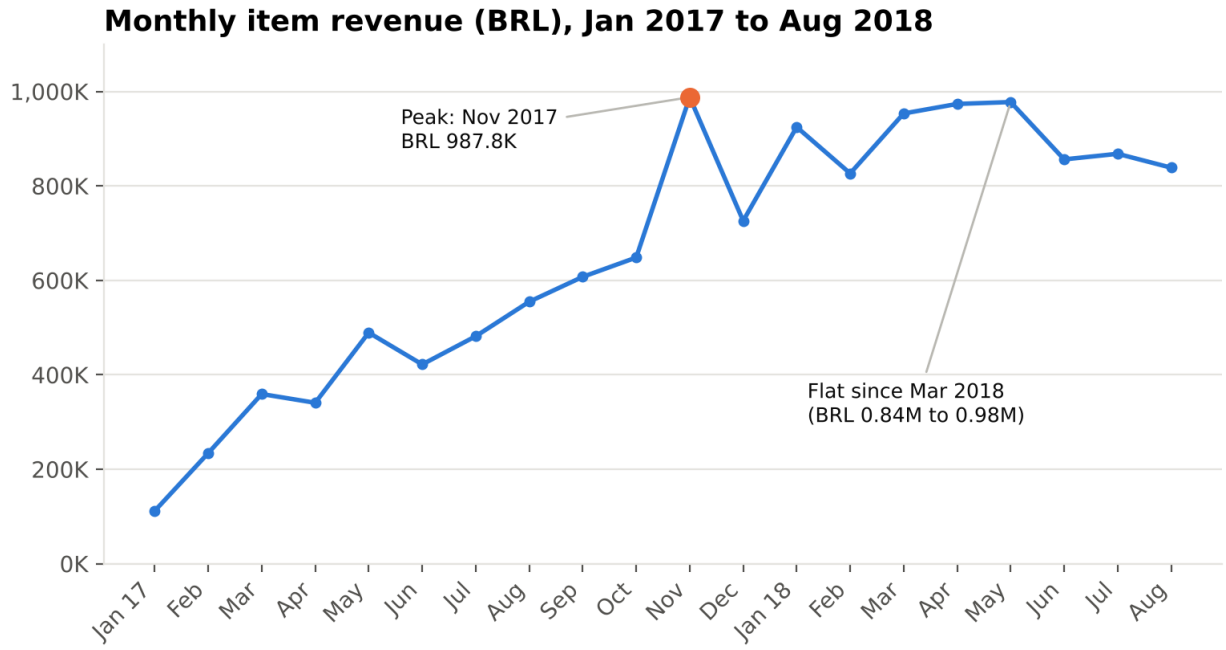
Answer: The best month was November 2017 (BRL 987,765.37) and the best complete quarter was Q2 2018 (BRL 2.81M). The weakest month was January 2017 (BRL 111,798.36), the first month of the window, and the weakest complete quarter was Q1 2017 (BRL 705,220.61).

Monthly item revenue

Month	Orders	Item revenue (BRL)
2017-01	750	111,798.36
2017-02	1,653	234,223.40
2017-03	2,546	359,198.85
2017-04	2,303	340,669.68
2017-05	3,546	489,338.25
2017-06	3,135	421,923.37
2017-07	3,872	481,604.52
2017-08	4,193	554,699.70
2017-09	4,150	607,399.67
2017-10	4,478	648,247.65
2017-11	7,289	987,765.37
2017-12	5,513	726,033.19
2018-01	7,069	924,645.00
2018-02	6,555	826,437.13
2018-03	7,003	953,356.25
2018-04	6,798	973,534.09
2018-05	6,749	977,544.69
2018-06	6,099	856,077.86
2018-07	6,159	867,953.46
2018-08	6,351	838,576.64

Quarterly item revenue

Quarter	Months in data	Orders	Item revenue (BRL)	vs previous quarter
2017 Q1	3	4,949	705,220.61	
2017 Q2	3	8,984	1,251,931.30	+77.5%
2017 Q3	3	12,215	1,643,703.89	+31.3%
2017 Q4	3	17,280	2,362,046.21	+43.7%
2018 Q1	3	20,627	2,704,438.38	+14.5%
2018 Q2	3	19,646	2,807,156.64	+3.8%
2018 Q3 (partial)	2	12,510	1,706,530.10	partial, not comparable



Source: SQL Server, dbo.v_qual_items (delivered orders only). Revenue = SUM(price), freight excluded.

Findings

- **Q3 2018 is partial** (July and August only), so it is excluded from the best and weakest quarter ranking.
- Quarter-over-quarter growth slowed from +77.5% (Q2 2017) to +3.8% (Q2 2018).
- The quarterly revenues add up to BRL 13,181,027.13, matching Q4.

Approach: Monthly and quarterly aggregates with GROUP BY, and RANK() OVER (ORDER BY revenue DESC/ASC) in CTEs to pick the best and weakest periods. HAVING COUNT(DISTINCT order_month) = 3 restricts the ranking to complete quarters.

Q6. Month-over-month revenue growth

Question: What is the month-over-month revenue growth rate? Use a CTE and LAG(), and handle the first month and zero denominators.

Answer: Month-over-month growth was volatile in 2017, with a November spike of +52.4% followed by a -26.5% correction in December. From April 2018 revenue has plateaued between BRL 0.84M and 0.98M a month and has never beaten the November 2017 peak.

Month-over-month growth

Month	Revenue (BRL)	Previous month (BRL)	MoM growth %
2017-01	111,798.36	NULL	NULL
2017-02	234,223.40	111,798.36	109.51
2017-03	359,198.85	234,223.40	53.36
2017-04	340,669.68	359,198.85	-5.16
2017-05	489,338.25	340,669.68	43.64
2017-06	421,923.37	489,338.25	-13.78
2017-07	481,604.52	421,923.37	14.14
2017-08	554,699.70	481,604.52	15.18

Month	Revenue (BRL)	Previous month (BRL)	MoM growth %
2017-09	607,399.67	554,699.70	9.50
2017-10	648,247.65	607,399.67	6.73
2017-11	987,765.37	648,247.65	52.37
2017-12	726,033.19	987,765.37	-26.50
2018-01	924,645.00	726,033.19	27.36
2018-02	826,437.13	924,645.00	-10.62
2018-03	953,356.25	826,437.13	15.36
2018-04	973,534.09	953,356.25	2.12
2018-05	977,544.69	973,534.09	0.41
2018-06	856,077.86	977,544.69	-12.43
2018-07	867,953.46	856,077.86	1.39
2018-08	838,576.64	867,953.46	-3.38

Findings

- **First month:** January 2017 has no previous month, so LAG() returns NULL and the growth is NULL. **Zero denominators:** NULLIF(prev_revenue, 0) returns NULL instead of an error (none occurred).
- **Largest increases:** Feb 2017 (+109.5%, tiny base), Mar 2017 (+53.4%), Nov 2017 (+52.4%). **Largest drops:** Dec 2017 (-26.5%), Jun 2017 (-13.8%), Jun 2018 (-12.4%).
- **November 2017:** orders rose 62.8% (4,478 to 7,289) but revenue rose only 52.4%, so average order value fell from BRL 144.76 to BRL 135.51. That fits a promotion-driven spike with smaller baskets, though the data cannot prove the cause.
- May 2018 (BRL 977,544.69) came within 1% of the peak but did not beat it.

Approach: Monthly revenue in a CTE, then LAG(revenue) OVER (ORDER BY order_month) in a second CTE, then (revenue - prev) / NULLIF(prev, 0).

Key SQL

```
WITH m AS (
    SELECT order_month, SUM(price) AS revenue
    FROM dbo.v_qual_items GROUP BY order_month
), g AS (
    SELECT order_month, revenue,
           LAG(revenue) OVER (ORDER BY order_month) AS prev_revenue
    FROM m
)
SELECT order_month, revenue, prev_revenue,
       CAST(100.0 * (revenue - prev_revenue)
            / NULLIF(prev_revenue, 0) AS DECIMAL(10,2)) AS mom_growth_pct
FROM g ORDER BY order_month;
```

Q7. Revenue and AOV by state

Question: Which Brazilian states generate the most revenue and the highest average order value? Does the highest-revenue region also have the highest AOV?

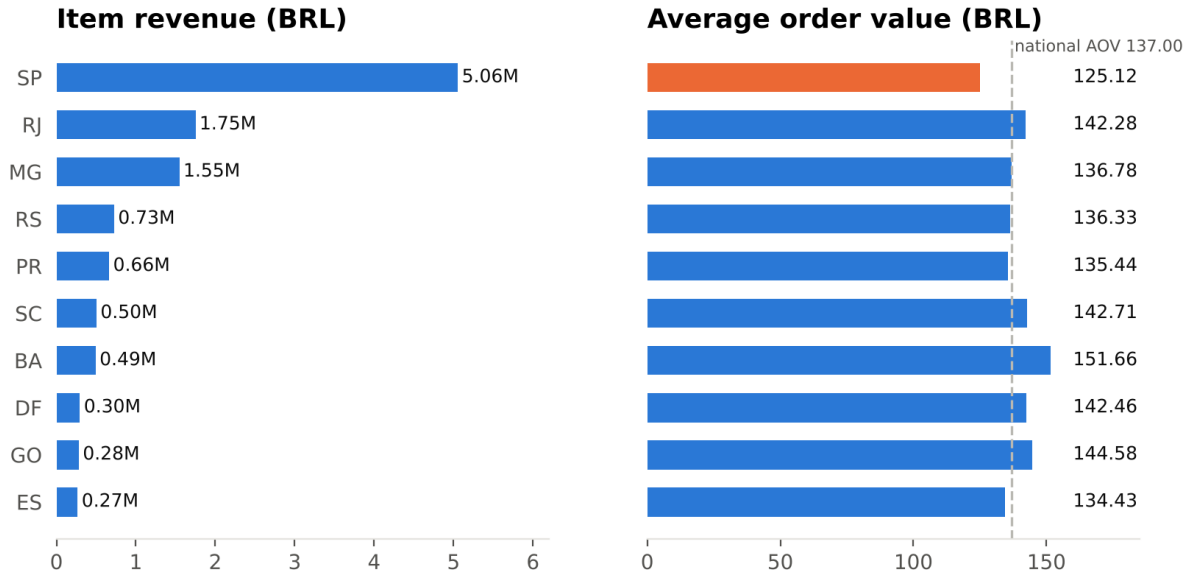
Answer: No. Sao Paulo (SP) generates the most revenue (BRL 5.06M, 38.4% of the total) but has the lowest AOV of all 27 states (BRL 125.12 vs BRL 137.00 nationally). The highest AOVs are in small northern and northeastern states.

Revenue and AOV by customer state (all 27)

State	Orders	Revenue (BRL)	% of revenue	AOV (BRL)	Revenue rank	AOV rank
SP	40,406	5,055,587.13	38.36	125.12	1	27
RJ	12,310	1,751,433.85	13.29	142.28	2	22
MG	11,319	1,548,206.88	11.75	136.78	3	23
RS	5,328	726,373.92	5.51	136.33	4	24
PR	4,903	664,048.00	5.04	135.44	5	25
SC	3,537	504,774.07	3.83	142.71	6	20
BA	3,253	493,339.26	3.74	151.66	7	18
DF	2,074	295,454.64	2.24	142.46	8	21
GO	1,950	281,932.21	2.14	144.58	9	19
ES	1,992	267,783.66	2.03	134.43	10	26
PE	1,587	250,639.89	1.90	157.93	11	16

State	Orders	Revenue (BRL)	% of revenue	AOV (BRL)	Revenue rank	AOV rank
CE	1,273	218,107.70	1.65	171.33	12	12
PA	942	173,382.99	1.32	184.06	13	6
MT	885	152,143.82	1.15	171.91	14	11
MA	713	116,305.12	0.88	163.12	15	15
MS	701	115,429.97	0.88	164.66	16	14
PB	516	112,536.92	0.85	218.09	17	1
PI	475	84,511.00	0.64	177.92	18	7
RN	470	81,376.97	0.62	173.14	19	10
AL	396	78,805.73	0.60	199.00	20	4
SE	332	56,288.74	0.43	169.54	21	13
TO	274	48,402.51	0.37	176.65	22	8
RO	243	45,682.76	0.35	187.99	23	5
AM	145	22,155.84	0.17	152.80	24	17
AC	80	15,930.97	0.12	199.14	25	3
AP	67	13,374.81	0.10	199.62	26	2
RR	40	7,017.77	0.05	175.44	27	9

Top 10 states: the biggest market (SP) has the lowest AOV of all 27 states



Source: SQL Server, Q7. SP AOV rank = 27 of 27.

Findings

- **Concentration:** SP, RJ and MG generate 63.4% of revenue; the top five states generate 74.0%.
- **Highest AOV:** PB (BRL 218.09), AP, AC, AL and RO (about BRL 188 to 218). These states have only 40 to 516 orders, and AC, AP and RR have fewer than 100, so their AOV is volatile.
- **Among large markets:** BA has the highest AOV of the top 10 states (BRL 151.66).
- The order counts add up to 96,211, matching Q4.

Approach: Order-level revenue (v_qual_order_rev) grouped by state, with RANK() OVER on revenue and on AOV and the share calculated with SUM() OVER ().

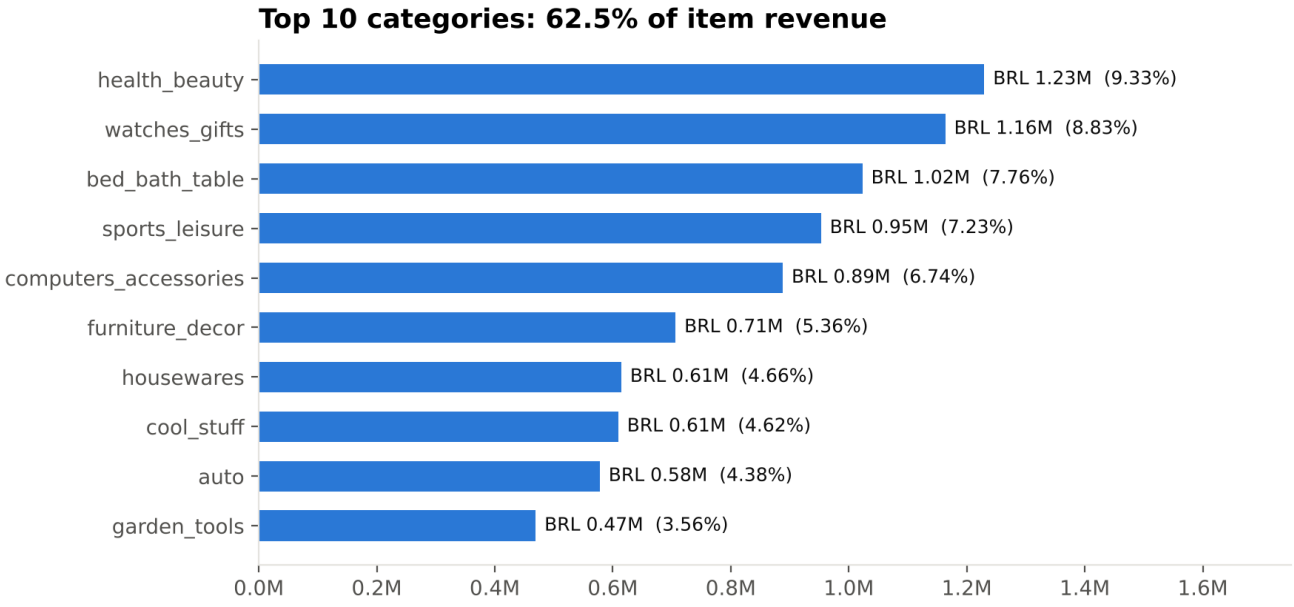
Q8. Top 10 categories by item revenue

Question: Which product categories generate the most item revenue? Return the top 10 with revenue, order counts and percentage contribution.

Answer: Ten categories generate 62.47% of item revenue, led by health_beauty (9.33%), watches_gifts (8.83%) and bed_bath_table (7.76%).

Top 10 categories

Category	Revenue (BRL)	Orders	Units	% of total	Cumulative %
health_beauty	1,229,557.50	8,610	9,422	9.33	9.33
watches_gifts	1,163,465.91	5,491	5,855	8.83	18.16
bed_bath_table	1,022,955.77	9,267	10,945	7.76	25.92
sports_leisure	952,840.40	7,513	8,414	7.23	33.14
computers_accessories	888,055.59	6,518	7,632	6.74	39.88
furniture_decor	706,237.17	6,258	8,095	5.36	45.24
housewares	614,341.62	5,734	6,783	4.66	49.90
cool_stuff	609,158.00	3,552	3,711	4.62	54.52
auto	577,838.39	3,802	4,132	4.38	58.91
garden_tools	469,135.40	3,443	4,263	3.56	62.47



Source: SQL Server, Q8. Share = % of total item revenue of BRL 13.18M.

Findings

- **Volume vs value:** bed_bath_table has the most orders (9,267) and units (10,945) but ranks third by revenue. watches_gifts ranks second by revenue on only 5,491 orders, at about BRL 199 per unit versus BRL 93 for bed_bath_table.
- The top 5 categories earn 39.88% of revenue.
- A category's order count is the number of orders containing it; an order with two categories counts in both, so the order column should not be added up.

Approach: Category revenue, distinct orders and units in a CTE; share and running total with SUM() OVER () and SUM() OVER (ORDER BY revenue DESC ROWS UNBOUNDED PRECEDING).

Q9. Products by units and by revenue

Question: Which products sell the most units, and which generate the most revenue? Compare the two rankings and explain why they differ.

Answer: Only 5 of the 15 leading products rank in the top 10 on both units and revenue. Unit leaders sell at BRL 22 to 150 each, while revenue leaders sell at BRL 118 to 1,392, so ranking by units alone would misdirect merchandising and inventory decisions.

Products in the top 10 by units or by revenue (IDs shortened)

Product	Category	Units	Revenue (BRL)	Avg price (BRL)	Units rank	Revenue rank
aca2eb7d...	furniture_decor	520	37,104.30	71.35	1	10
422879e1...	garden_tools	484	26,577.22	54.91	2	17

Product	Category	Units	Revenue (BRL)	Avg price (BRL)	Units rank	Revenue rank
99a4788c...	bed_bath_table	477	42,049.66	88.15	3	5
389d119b...	garden_tools	390	21,336.79	54.71	4	24
368c6c73...	garden_tools	388	21,056.80	54.27	5	25
53759a2e...	garden_tools	373	20,387.20	54.66	6	26
d1c42706...	computers_accessories	332	45,620.56	137.41	7	4
53b36df6...	watches_gifts	321	37,454.64	116.68	8	9
154e7e31...	health_beauty	274	6,173.26	22.53	9	213
3dd2a171...	computers_accessories	272	40,782.80	149.94	10	6
bb50f2e2...	health_beauty	194	63,560.00	327.63	14	1
53c63843...	health_beauty	153	53,652.30	350.67	23	2
5f504b3a...	cool_stuff	63	37,733.90	598.95	114	8
25c38557...	baby	38	38,907.32	1,023.88	259	7
d6160fb7...	computers	33	45,949.35	1,392.40	314	3

Findings

- **Why they differ:** revenue = units x price. Cheap products win on units (a health_beauty item sold 274 units at BRL 22.53 and ranks 9th by units but 213th by revenue); expensive products win on revenue (a computers item sold 33 units at BRL 1,392.40 and ranks 3rd by revenue but 314th by units).
- **No single product dominates:** the top product by revenue (BRL 63,560.00) is 0.48% of total revenue, and the top by units (520) is 0.47% of units sold. The concentration is at category level, not product level.

Approach: Product-level units and revenue in a CTE, then two RANK() OVER windows (units and revenue); the filter keeps products ranked in the top 10 on either measure.

Q10. High-volume categories with weak reviews or slow delivery

Question: Which categories have high sales volume but relatively weak review scores or poor delivery outcomes? Check the grain of each table before joining so review counts and revenue are not inflated.

Answer: Five of the 20 highest-volume categories are flagged. office_furniture combines the weakest rating (3.65) with the slowest delivery (20.5 days). bed_bath_table, the largest category by orders, has weak reviews (4.00 stars, 16.04% 1-2 star) without a delivery problem.

Grain check: joining raw reviews inflates revenue

Join	Revenue (BRL)
Items only (correct)	13,181,027.13
After joining raw order_reviews	13,239,238.92 (+58,211.79, +0.44%)
After joining deduplicated reviews (v_reviews_dedup)	13,181,027.13

20 highest-volume categories (benchmarks: 4.16 stars, 6.79% late)

Rank	Category	Orders	Revenue (BRL)	Avg review	% 1-2 star	Avg days	Late %	Flag
1	bed_bath_table	9,267	1,022,955.77	4.00	16.04	12.9	7.43	Weak reviews
2	health_beauty	8,610	1,229,557.50	4.23	11.38	12.0	7.51	OK
3	sports_leisure	7,513	952,840.40	4.23	11.39	12.2	6.59	OK
4	computers_accessories	6,518	888,055.59	4.08	14.54	13.1	6.38	OK
5	furniture_decor	6,258	706,237.17	4.06	15.33	13.0	7.17	OK
6	housewares	5,734	614,341.62	4.19	11.79	11.0	5.37	OK
7	watches_gifts	5,491	1,163,465.91	4.12	13.71	12.7	7.40	OK
8	telephony	4,086	309,631.94	4.05	14.13	12.9	7.12	Weak reviews
9	auto	3,802	577,838.39	4.15	12.98	12.2	7.31	OK
10	toys	3,782	467,179.98	4.24	11.38	11.6	6.43	OK
11	cool_stuff	3,552	609,158.00	4.22	11.27	12.3	5.91	OK
12	garden_tools	3,443	469,135.40	4.18	12.24	13.6	6.53	OK
13	perfumery	3,060	385,315.55	4.26	11.80	11.8	6.60	OK
14	baby	2,800	399,050.67	4.10	14.25	12.6	8.07	Late deliveries
15	electronics	2,516	154,935.94	4.12	13.22	12.8	7.63	OK
16	stationery	2,264	223,788.69	4.30	10.04	12.7	6.54	OK

Rank	Category	Orders	Revenue (BRL)	Avg review	% 1-2 star	Avg days	Late %	Flag
17	fashion_bags_accessories	1,812	148,945.19	4.24	10.09	11.2	5.63	OK
18	pet_shop	1,684	211,005.96	4.27	10.68	11.3	5.11	OK
19	unknown	1,390	170,660.74	4.01	17.54	12.7	7.27	Weak reviews
20	office_furniture	1,248	266,853.43	3.65	21.71	20.5	8.09	Weak reviews + late*

Findings

- **Flag rule:** weak = at least 0.10 stars below the 4.16 benchmark; late = at least 1 point above the 6.79% benchmark. *The flag text for office_furniture was truncated in the SQL output; applying the rule (3.65 <= 4.06 and 8.09 >= 7.79) gives "Weak reviews + late deliveries".
- **office_furniture is the clearest problem:** 0.51 stars below benchmark, 21.71% of reviews are 1-2 stars, and 20.5 days to deliver (about 64% longer than the 12.48-day average).
- **bed_bath_table is the most valuable problem:** top category by orders and 7.76% of revenue, but 0.16 stars below benchmark with 16.04% 1-2 star reviews (health_beauty and sports_leisure sit near 11.4%). Its delivery time (12.9 days) and late rate (7.43%) are normal, so delivery does not explain it; the cause is more likely product quality or customer expectations, which the data cannot separate.
- **Watch list:** furniture_decor (4.06 stars, 15.33% 1-2 star) and computers_accessories (4.08, 14.54%) sit just above the flag threshold.
- **unknown** is a data gap (610 products with no category), not a real category.
- bed_bath_table, telephony and office_furniture together hold BRL 1.60M, or 12.1% of revenue.

Approach: Aggregated to one row per order x category first, then joined to one review per order (v_reviews_dedup) and one delivery record per order, so neither revenue nor review counts can be inflated. Benchmarks come from a separate CTE and a CASE expression sets the flag.

Key SQL

```
-- One review per order (latest wins), so joins cannot inflate revenue
CREATE OR ALTER VIEW dbo.v_reviews_dedup AS
SELECT order_id, review_score, review_creation_date, review_answer_timestamp
FROM (
    SELECT order_id, review_score, review_creation_date, review_answer_timestamp,
           ROW_NUMBER() OVER (PARTITION BY order_id
                              ORDER BY review_creation_date DESC,
                                       review_answer_timestamp DESC) AS rn
    FROM dbo.order_reviews
) x
WHERE rn = 1;
```

Part D: Customer intelligence

Q11. Repeat purchasers

Question: What percentage of customers are repeat purchasers? State the definition and the denominator.

Answer: Only 3.00% of customers (2,789 of 93,104) placed more than one qualifying order, falling to 2.15% when orders placed on the same day are excluded. 97% of customers bought once.

Repeat purchase rate

Measure	Customers	% of 93,104
Repeat purchasers (2 or more orders)	2,789	3.00
Stricter: orders on different days	2,005	2.15

Orders per customer

Orders placed	Customers	% of customers
1	90,315	97.00
2	2,562	2.75
3	180	0.19
4	28	0.03
5	9	0.01
6	5	0.01
7	3	0.00
9	1	0.00
15	1	0.00

Findings

- **Definition:** a customer (`customer_unique_id`) with more than one qualifying order. **Denominator:** all 93,104 customers with at least one qualifying order.
- **Most repeaters stop at two orders:** 2,562 of the 2,789 (91.9%). Only 227 customers placed three or more.
- **784 of the 2,789 repeaters (28.1%) ordered only on a single day.** These are probably one basket split into several order IDs, not a genuine return visit, which is why 2.15% is the conservative figure.
- The window is 20 months and ends in Aug 2018, so a customer who would have returned later is not counted.

Approach: COUNT(DISTINCT order_id) and COUNT(DISTINCT CAST(order_purchase_timestamp AS DATE)) per customer_unique_id in a CTE, then conditional counts.

Q12. Top 20 customers by spend

Question: Who are the top 20 customers by total item spend? Show order counts and AOV. The identifiers are anonymised; do not try to identify individuals.

Answer: The 20 highest-spending customers spent BRL 107,712.36, only 0.82% of revenue, and 18 of them placed a single order. High spend comes from occasional large purchases, not from loyal heavy buyers.

Top 20 customers (anonymised IDs shortened to 8 characters)

Rank	Customer	Orders	Total spend (BRL)	AOV (BRL)
1	0a0a9211...	1	13,440.00	13,440.00
2	da122df9...	2	7,388.00	3,694.00
3	763c8b1c...	1	7,160.00	7,160.00
4	dc4802a7...	1	6,735.00	6,735.00
5	459bef48...	1	6,729.00	6,729.00
6	ff4159b9...	1	6,499.00	6,499.00
7	4007669d...	1	5,934.60	5,934.60
8	eebb5dda...	1	4,690.00	4,690.00
9	48e1ac10...	1	4,590.00	4,590.00
10	a229eba7...	1	4,400.00	4,400.00
11	edde2314...	1	4,399.87	4,399.87

Rank	Customer	Orders	Total spend (BRL)	AOV (BRL)
12	fa562ef2...	1	4,099.99	4,099.99
13	c8460e42...	4	4,080.00	1,020.00
14	ca27f3da...	1	4,059.00	4,059.00
15	011875f0...	1	3,999.90	3,999.90
16	edf81e1f...	1	3,999.00	3,999.00
17	5e713be0...	1	3,980.00	3,980.00
18	f0767ae7...	1	3,930.00	3,930.00
19	58483a1c...	1	3,899.00	3,899.00
20	3fec1a05...	1	3,700.00	3,700.00

Findings

- **The top 20 are 0.817% of total revenue**, so there is no dependence on a handful of big accounts.
- **One-off big purchases:** only rank 2 (2 orders) and rank 13 (4 orders, BRL 1,020 average) placed more than one order.
- **The #1 order is 98 times the average order value** (BRL 13,440 vs BRL 137.00). It exceeds the highest single item price in the data (BRL 6,735), so it contained several items.
- Even rank 20 (BRL 3,700) is 27 times the average order. IDs are anonymised hashes and no attempt is made to identify anyone.

Approach: ROW_NUMBER() OVER (ORDER BY SUM(revenue) DESC) on order-level revenue grouped by customer_unique_id; the share of the top 20 uses ROW_NUMBER() with a customer-id tie-breaker.

Q13. RFM segmentation

Question: Can customers be segmented using RFM analysis? Calculate recency, frequency and monetary value, score with NTILE(), create named segments, and explain the scoring rules and the limitations of a historical dataset.

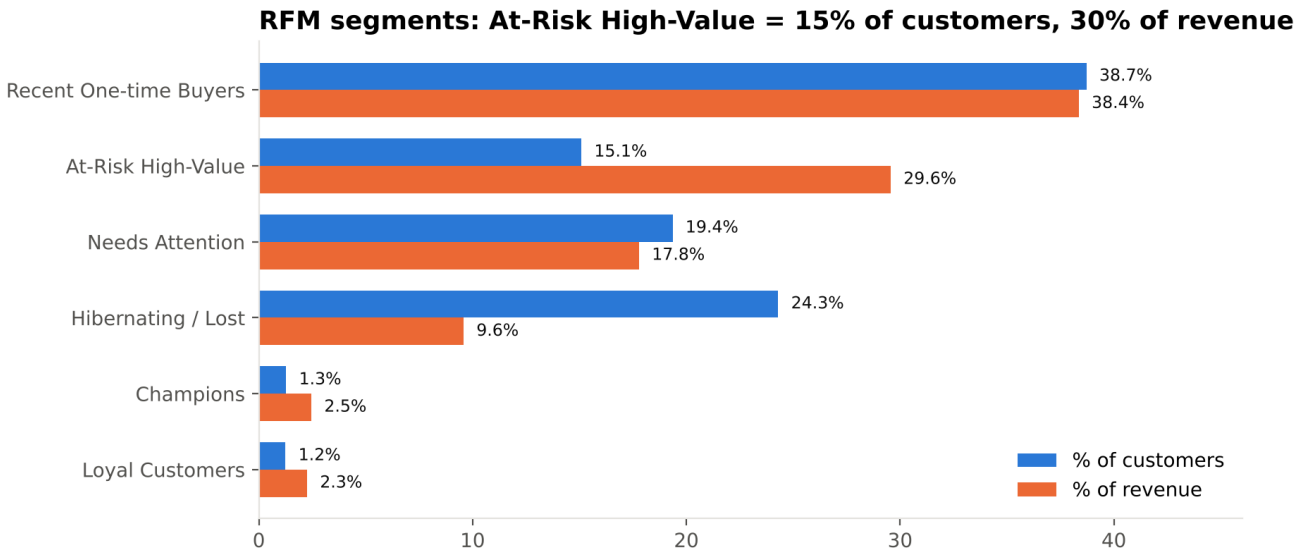
Answer: Yes. Two groups of one-time buyers hold 67.9% of revenue: 36,046 recent buyers (BRL 5.06M) and 14,053 lapsed high spenders (BRL 3.90M, about BRL 277 each). Only 2.5% of customers are repeat buyers in the Champions or Loyal segments.

Segment summary

Segment	Customers	% cust.	Revenue (BRL)	% rev.	Avg recency (days)	Avg orders	Avg spend (BRL)
Recent One-time Buyers	36,046	38.72	5,055,929.41	38.36	90.7	1.00	140.26
At-Risk High-Value Customers	14,053	15.09	3,895,867.54	29.56	393.7	1.02	277.23
Needs Attention	18,023	19.36	2,344,788.93	17.79	219.9	1.00	130.10
Hibernating / Lost	22,626	24.30	1,263,084.62	9.58	393.7	1.01	55.82
Champions	1,195	1.28	323,876.74	2.46	88.8	2.15	271.03
Loyal Customers	1,161	1.25	297,479.89	2.26	264.8	2.09	256.23

Scoring bands (NTILE(5) for recency and monetary; frequency by rule)

Score	Recency (days since last order)	Monetary (total spend, BRL)	Frequency (orders)
5	1 - 92	179.90 - 13,440.00	3 or more
4	92 - 177	109.90 - 179.90	
3	177 - 268	69.90 - 109.90	2
2	268 - 381	39.90 - 69.90	
1	381 - 602	0.85 - 39.90	1



Source: SQL Server, Q13 (dbo.v_rfm). R and M scored with NTILE(5); F by rule.

Findings

- **Reference date:** the day after the last qualifying order (end of Aug 2018), not today, so recency is measured inside the dataset's own timeline.
- **Scoring:** Recency and Monetary use NTILE(5), about 18,621 customers per band (5 = most recent / highest spend). Frequency is rule-based (1 order = 1, 2 orders = 3, 3 or more = 5), because 97% of customers have one order and NTILE would split identical values arbitrarily. A customer-id tie-breaker makes the split reproducible, which is why a band edge such as BRL 39.90 appears in two bands.
- **Segment rules (first match wins):** Champions = R >= 4 and F >= 3; Loyal = F >= 3 and R >= 2; At-Risk High-Value = R <= 2 and M >= 4; Recent One-time Buyers = R >= 4; Needs Attention = R = 3; everything else = Hibernating / Lost.
- **At-Risk High-Value is the segment to talk about:** 15% of customers but 29.6% of revenue, spending about twice the average customer (BRL 277.23 vs BRL 141.57). With 1.02 orders on average these are lapsed one-time big spenders, not former loyalists.
- **Hibernating / Lost** is 24.3% of customers but only 9.6% of revenue (BRL 55.82 average spend), so it is low priority. The six segments add up to BRL 13,181,027.13, matching Q4.
- **Limitations:** (1) recency is measured against the dataset end date, not today; (2) "at risk" is a recency label, not a modelled churn probability; (3) frequency separates only 3% of customers; (4) there are no customer attributes such as age or channel; (5) IDs are anonymised; the segments are descriptive, not predictive.

Approach: A view (v_rfm) computes recency (DATEDIFF), frequency and monetary per customer_unique_id, scores them with NTILE(5) OVER (ORDER BY ...), and assigns segments with a CASE expression.

Key SQL

```
scored AS (
    SELECT *,
           NTILE(5) OVER (ORDER BY recency_days DESC, customer_unique_id) AS r_score,
           CASE WHEN frequency >= 3 THEN 5 WHEN frequency = 2 THEN 3 ELSE 1 END AS f_score,
           NTILE(5) OVER (ORDER BY monetary, customer_unique_id)          AS m_score
    FROM cust
)
SELECT *,
       CASE WHEN r_score >= 4 AND f_score >= 3 THEN 'Champions'
            WHEN f_score >= 3 AND r_score >= 2 THEN 'Loyal Customers'
            WHEN r_score <= 2 AND m_score >= 4 THEN 'At-Risk High-Value Customers'
            WHEN r_score >= 4                  THEN 'Recent One-time Buyers'
            WHEN r_score = 3                   THEN 'Needs Attention'
            ELSE 'Hibernating / Lost' END AS segment
FROM scored;
```

Part E: Operations and executive recommendations

Q14. Delivery performance against the estimated date

Question: How well is the business meeting estimated delivery dates? Calculate the on-time percentage, average delivery duration and late-delivery rate, and investigate differences across states or categories.

Answer: 93.21% of delivered orders arrived on or before the estimated date (late rate 6.79%), with an average delivery time of 12.48 days against 24.28 promised. Late rates range from 4.50% in Sao Paulo to 21.46% in Alagoas, and three months (Nov 2017, Feb and Mar 2018) account for about 48% of all late orders.

Overall delivery performance

Metric	Value
Delivered orders analysed	96,203 (96,211 qualifying minus 8 with no delivery date)
On time or early	89,672 (93.21%)
Late	6,531 (6.79%)
Average delivery time	12.48 days
Average promised time	24.28 days
Average lateness when late	10.62 days

By state (24 states with at least 100 delivered orders)

State	Delivered orders	Avg delivery days	Late %
AL	396	24.48	21.46
MA	713	21.47	17.53
SE	332	21.38	15.36
PI	475	19.38	13.89
CE	1,273	21.17	13.83
BA	3,253	19.28	12.17
RJ	12,310	15.22	12.14
PA	942	23.71	11.25
ES	1,992	15.71	10.74
PB	516	20.36	10.47
TO	274	17.60	9.85
MS	701	15.54	9.70
PE	1,587	18.40	9.64
RN	470	19.16	9.36
SC	3,537	14.89	8.23
GO	1,950	15.54	6.56
RS	5,327	15.24	6.10
MT	885	18.02	5.99
DF	2,074	12.87	5.69
MG	11,319	11.92	4.59
SP	40,399	8.68	4.50
PR	4,903	11.92	4.06
RO	243	19.28	2.88
AM	145	26.36	2.76

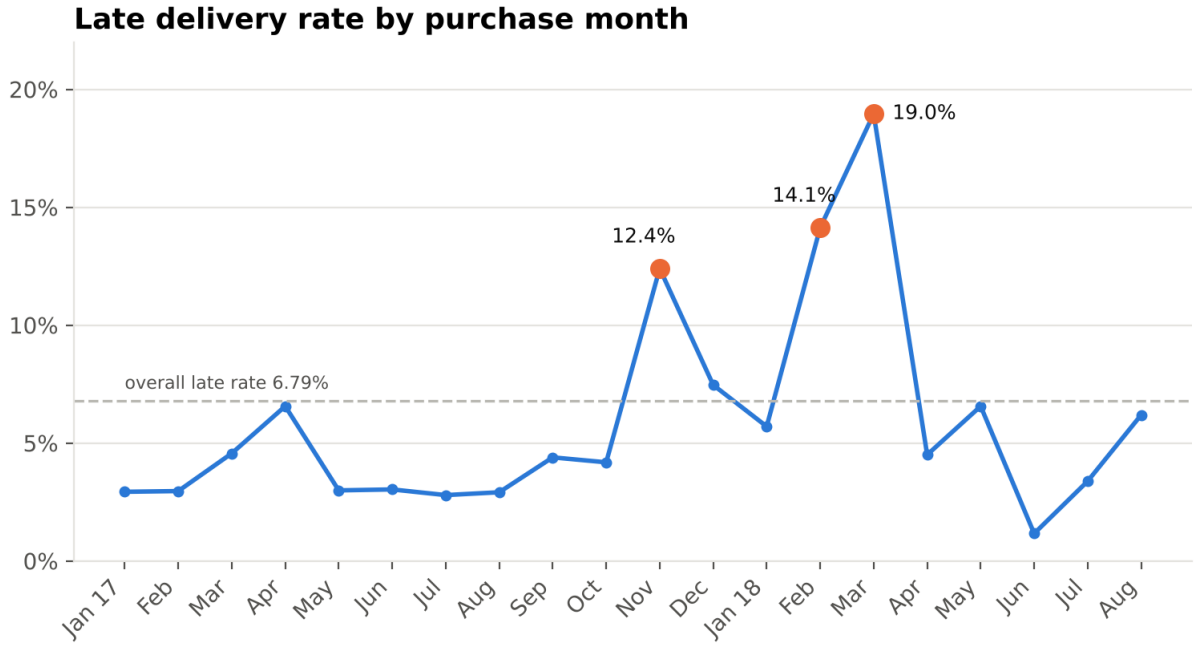
By category (32 categories with at least 300 delivered orders)

Category	Delivered orders	Avg delivery days	Late %
audio	346	13.44	11.85
home_confort	392	13.64	9.44
office_furniture	1,248	20.50	8.09
baby	2,800	12.60	8.07
electronics	2,516	12.84	7.63
musical_instruments	611	13.02	7.53
health_beauty	8,610	11.97	7.51
furniture_living_room	414	13.84	7.49

Showing the first 8 of 32 rows. The full table is in BUSINESS_QUESTIONS_AND_ANSWERS.md.

By purchase month

Month	Delivered orders	Late %
2017-01	750	2.93
2017-02	1,653	2.96
2017-03	2,546	4.56
2017-04	2,303	6.56
2017-05	3,545	2.99
2017-06	3,135	3.03
2017-07	3,872	2.79
2017-08	4,193	2.91
2017-09	4,150	4.39
2017-10	4,478	4.18
2017-11	7,288	12.40
2017-12	5,513	7.46
2018-01	7,069	5.70
2018-02	6,555	14.13
2018-03	7,003	18.96
2018-04	6,798	4.50
2018-05	6,749	6.56
2018-06	6,096	1.16
2018-07	6,156	3.38
2018-08	6,351	6.19



Source: SQL Server, Q14. Late = delivered date after estimated date.

Findings

- **On time** = delivered date on or before the estimated date, compared at date level. **Late** = delivered date after the estimated date.
- **The estimates are generous:** orders arrive on average 11.8 days ahead of the promised date. The 6.79% that are late are the tail, and when an order is late it is late by a lot (10.62 days).
- **Remote states are slow and late:** northern and northeastern states take 19 to 26 days versus 8.7 days in SP.
- **RJ stands out:** the second-largest state (12,310 orders, about 1,490 late orders calculated from its rate) has the same average delivery time as RS (15.2 days) but twice the late rate (12.14% vs 6.10%). Slow transit alone does not explain it; either promised dates are too tight there or delivery times vary widely, which the data cannot tell apart.

- **Categories vary less than states:** late rates run from 4.32% (luggage_accessories) to 11.85% (audio), a 7.5-point spread against 18.7 points across states, so where a customer lives matters more than what they buy. office_furniture has the longest delivery time (20.5 days).
- **Three months stand out:** Nov 2017 (12.40%), Feb 2018 (14.13%) and Mar 2018 (18.96%) produced about 48% of all late orders (calculated from the monthly rates and order counts). November coincides with the volume peak; February and March 2018 do not, because their volumes were no higher than January, so volume alone does not explain them. After March the late rate fell to 1.16% in June, then rose to 6.19% in August.

Approach: Late-delivery flag computed once in v_qual_orders (is_late_day, date-level comparison); DATEDIFF for delivery and promised durations; GROUP BY state, category and month with HAVING to keep only groups with enough orders.

Key SQL

```
SELECT COUNT(*) AS delivered_orders,
       CAST(100.0 * SUM(CASE WHEN is_late_day = 0 THEN 1 ELSE 0 END)
            / COUNT(*) AS DECIMAL(5,2)) AS on_time_pct,
       CAST(AVG(CAST(delivery_days AS FLOAT)) AS DECIMAL(6,2)) AS avg_delivery_days
FROM dbo.v_qual_orders
WHERE is_delivered_valid = 1;
```

Q15. Three evidence-based business opportunities

Question: What are the three most important business opportunities revealed by the data? Each recommendation must reference a measurable result from the earlier SQL outputs.

Answer: (1) Reduce late deliveries in the worst states and peak months; (2) convert one-time buyers into repeat customers; (3) fix review quality in the large flagged categories. Each is tied to a measured result below.

Evidence A: repeat vs one-time customers

Customer type	Customers	% customers	Revenue (BRL)	% revenue	Avg spend (BRL)
One-time	90,315	97.00	12,455,672.09	94.50	137.91
Repeat	2,789	3.00	725,355.04	5.50	260.08

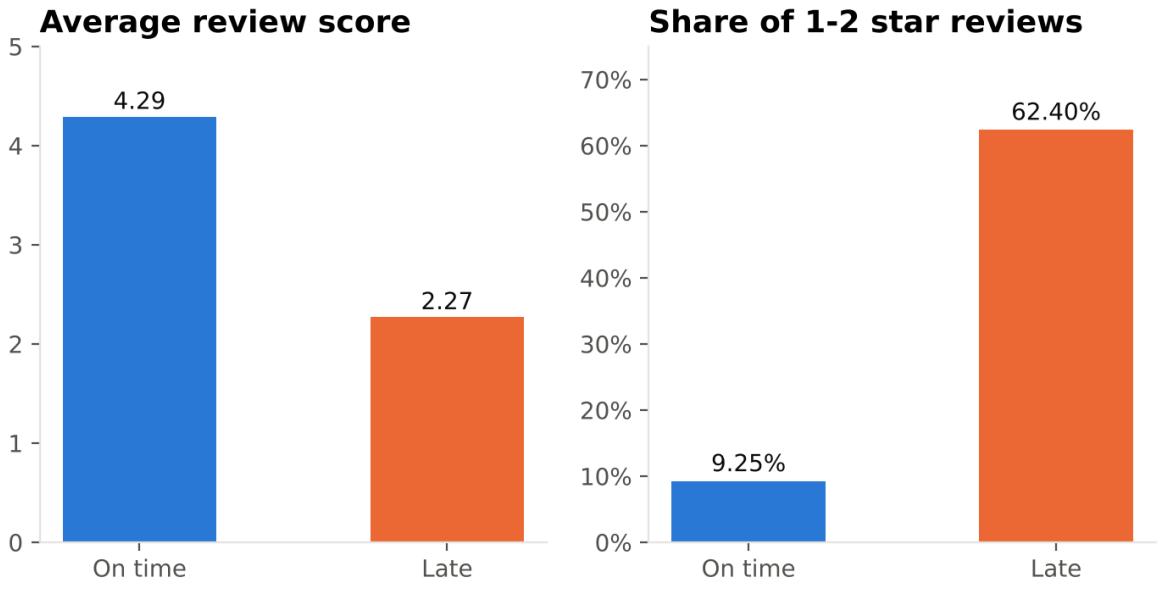
Evidence B: delivery vs review score

Delivery	Orders	Avg review	% 1-2 star
Late	6,378	2.27	62.40
On time	89,182	4.29	9.25

Evidence C: revenue concentration

Measure	Value
Top 5 categories, share of revenue	39.88%
Top 10 categories, share of revenue	62.47%
SP, RJ, MG, share of revenue (Q7)	63.4%

Late deliveries cut the average review score from 4.29 to 2.27



Source: SQL Server, Q15 (evidence B). On time: 89,182 orders; late: 6,378 orders.

Findings

- **1. Cut late deliveries in the worst states and peak months.** Late orders average **2.27 stars vs 4.29**, and **62.40% vs 9.25%** of them get 1-2 stars. Late orders are 6.8% of delivered orders but generate about a third (about 32.5%) of all 1-2 star reviews (calculated from the rates and counts above). The problem is concentrated: AL has a 21.46% late rate, MA 17.53%, RJ 12.14% on 12,310 orders, and Nov 2017 with Feb to Mar 2018 produced about 48% of late orders. **Action:** set carrier service levels and capacity plans for peak months, run a root-cause review of Feb to Mar 2018, review promised dates and carriers in RJ, BA, MA, SE and AL, and warn customers proactively about predicted delays. **Measure:** bringing the national late rate from 6.79% to São Paulo's 4.50% means about 2,200 fewer late orders over the 20 months; if those orders earned on-time review profiles, that is about 1,170 fewer 1-2 star reviews (about 10% of the total). This is illustrative and assumes lateness causes the review gap; the data shows only a correlation.
- **2. Convert one-time buyers into repeat customers.** 97.00% of customers buy once, and repeaters are only 3.00% (2.15% on different days). Repeat customers spend **BRL 260.08 vs BRL 137.91**, 1.89 times a one-time customer. 14,053 At-Risk High-Value customers hold 29.56% of revenue (BRL 3.90M, about BRL 277 each) and 36,046 Recent One-time Buyers are still inside the window to repeat. Growth has also stalled: quarterly growth fell to +3.8% in Q2 2018 and monthly revenue has been flat since March 2018. **Action:** run a second-purchase offer in the first 30 to 90 days and a win-back campaign for the At-Risk High-Value segment, and track the repeat rate monthly. **Measure:** each extra percentage point of repeat rate is about 931 customers; at the observed repeat vs one-time spend gap (BRL 122.17) that is roughly BRL 114K, or +0.9% of revenue. This is illustrative and assumes converted customers behave like today's repeaters.
- **3. Fix review quality in the large flagged categories.** bed_bath_table is the #1 category by orders (9,267) and 7.76% of revenue, but rates 4.00 vs the 4.16 benchmark with 16.04% 1-2 star reviews (about 11.4% in comparable categories), and its delivery metrics are normal. office_furniture rates 3.65 with 21.71% 1-2 star reviews, a 20.5-day delivery time and an 8.09% late rate; telephony rates 4.05 with 14.13% 1-2 star reviews. Together these three categories hold **BRL 1.60M (12.1% of revenue)**, and the top 10 categories hold 62.47% of revenue, so quality problems in a few categories reach a large share of sales. **Action:** audit sellers and products in bed_bath_table (review score by seller) and coach or remove the weakest, fix bulky-goods logistics and delivery promises for office_furniture, and repair the missing category data (610 products). **Measure:** raise bed_bath_table from 4.00 to the 4.16 benchmark and cut its 1-2 star share from 16.04% toward 11.4%.
- **Also worth a sentence:** revenue is concentrated in SP, RJ and MG (63.4%), and SP has the lowest AOV of all 27 states (BRL 125.12 vs BRL 137.00 nationally).

Approach: Evidence queries reuse the clean views: customer-level order counts and spend, delivery flag joined to one review per order, and category revenue ranked with ROW_NUMBER().

Limitations

- Recency in the RFM analysis is measured against the dataset end date (Aug 2018), not today.
- "At-Risk" is a recency label, not a modelled churn probability.
- Findings such as late delivery and low reviews are correlations; the data cannot prove cause.
- Small states have unstable averages (for example AOV in states with under 100 orders).
- The dataset has no customer attributes and the identifiers are anonymised; no attempt was made to identify individuals.
- The analysis window (Jan 2017 to Aug 2018) leaves out 2016 and Sep-Oct 2018 on purpose.
- The sizing estimates in Q15 are illustrative assumptions, not forecasts.

Dataset: Brazilian E-Commerce Public Dataset by Olist (Kaggle). Identifiers are anonymised.